FOR A SMOOTH QUASI-COMPACT SCHEME WITH AFFINE DIAGONAL THE TWO INTERSECTION PRODUCTS AGREE AFTER TENSORING WITH THE RATIONALS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FFF (sources/stacks-chow/pinned-original-source.tex line 14587).

Allowed standard prerequisites (editorial).

- The intersection product built from the Chern character in the regular section. Same chapter, Section labelled section-intersection-regular, invoked at proof lines 14595-14601. Fixes one of the two products being compared.
- The identification of the bivariant ring with Chow homology for a smooth scheme. Same chapter, Lemma labelled lemma-identify-chow-for-smooth, cited at proof lines 14604-14606. Turns the cap c cap c' cap [X] into the smooth intersection product alpha . beta.

Source reference. Original section 0FEX: sources/stacks-chow/pinned-original-source.tex, source lines 13290–13290. Same-chapter pointer into the author environment lemma at source line 13290; invoked at proof line 14599 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FC2: sources/stacks-chow/pinned-original-source.tex, source lines 14403–14403. Same-chapter pointer into the author environment lemma at source line 14403; invoked at proof line 14603 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let k be a field. Let X be a smooth scheme over k which is quasi-compact and has affine diagonal. Then the intersection product on $\mathrm{CH}^*(X)$ constructed in this section agrees after tensoring with $\mathbf{Q}$ with the intersection product constructed in Section 0FEX.*

Proof. Let $\alpha \in \mathrm{CH}^i(X)$ and $\beta \in \mathrm{CH}^j(X)$. Write $\alpha = ch(\alpha') \cap [X]$ and $\beta = ch(\beta') \cap [X]$ $\alpha', \beta' \in K_0(\mathit{Vect}(X)) \otimes \mathbf{Q}$ as in Section 0FEX. Set $c = ch(\alpha')$ and $c' = ch(\beta')$. Then the intersection product in Section 0FEX produces $c \cap c' \cap [X]$. This is the

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same as $\alpha \cdot \beta$ by Lemma 0FC2 (or rather the generalization that $A^i(X) \to \mathrm{CH}^i(X)$, $c \mapsto c \cap [X]$ is an isomorphism for any smooth scheme X over k). $\square$